

Lab Project 02: Student Record Query Project

Target Audience: CSE Students Environment: XAMPP, phpMyAdmin, MySQL

1. Project Overview & Objectives

This lab project focuses on data manipulation and querying techniques using MySQL. Students will learn how to insert records into a relational database and perform data retrieval using fundamental SQL clauses.

- Insert structured student data into MySQL tables using DML (Data Manipulation Language).
- Retrieve complete and conditional datasets using `SELECT` and `WHERE` clauses.
- Sort retrieved datasets in ascending and descending order using `ORDER BY`.
- Verify data manipulation outcomes via the phpMyAdmin web interface.

2. Database Initialization & Schema Setup

Run the following script in phpMyAdmin SQL console to initialize the database and construct the `Students` entity schema.

```
CREATE DATABASE StudentDB;
USE StudentDB;

CREATE TABLE Students (
    student_id INT PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    department VARCHAR(50),
    cgpa DECIMAL(3,2),
    enrollment_year INT
);
```

3. Data Insertion (Manipulation)

Insert representative student records into the `Students` table for query testing.

```
INSERT INTO Students (student_id, first_name, last_name, department, cgpa,
enrollment_year) VALUES
(101, 'Rahim', 'Uddin', 'CSE', 3.75, 2023),
(102, 'Fatema', 'Nahar', 'EEE', 3.85, 2022),
(103, 'Tanvir', 'Hossain', 'CSE', 3.20, 2023),
(104, 'Nusrat', 'Jahan', 'BBA', 3.90, 2021),
(105, 'Arik', 'Chowdhury', 'CSE', 3.60, 2022);
```

4. Data Retrieval & Query Operations

4.1 Basic Data Retrieval

Retrieve all records and attributes from the `Students` table.

```
SELECT * FROM Students;
```

4.2 Filtering Records with WHERE Clause

Filter student records belonging strictly to the 'CSE' department with a CGPA greater than or equal to 3.50.

```
SELECT student_id, first_name, last_name, cgpa
FROM Students
WHERE department = 'CSE' AND cgpa >= 3.50;
```

4.3 Sorting Results with ORDER BY Clause

Retrieve all students and sort the result set by CGPA in descending order (highest to lowest).

```
SELECT * FROM Students
ORDER BY cgpa DESC;
```

4.4 Combining WHERE and ORDER BY Clauses

Find all CSE department students and order them by enrollment year in ascending order.

```
SELECT student_id, first_name, department, enrollment_year
FROM Students
WHERE department = 'CSE'
ORDER BY enrollment_year ASC;
```

Execution Note: Ensure that when combining clauses, `WHERE` always precedes `ORDER BY` in the SQL execution sequence to avoid syntax errors.

5. Submission Requirements

- SQL Source File containing schema creation, data insertion, and all four test queries.
- Screen captures demonstrating the query execution results in phpMyAdmin for each step.
- Package all artifacts into `StudentID_DBMS_Project02.zip` and submit via the university LMS portal.